

EXP. NO: 6

DATA VISUALIZATION USING POWER BI

AIM:

To learn the Power BI Desktop interface and workflow by connecting to Excel and CSV data sources, performing data cleaning and transformation using Power Query, creating calculated columns and DAX measures, and developing an interactive dashboard using different visualizations.

PROCEDURE:

1. Open Power BI Desktop and Explore the Interface

- a. Launch Power BI Desktop.
- b. Identify the Fields/Data, Visualizations, Filters, Report, Data, and Model views.
- c. Open the Report view to create the dashboard.

2. Connect to Data Sources

- a. Excel Data:
 - i. Select Home → Get Data → Excel Workbook.
 - ii. Select the SalesData.xlsx file.
 - iii. In the Navigator window, select the Sales sheet.
 - iv. Click Load.
- b. CSV Data:
 - i. Select Home → Get Data → Text/CSV.
 - ii. Select the Targets.csv file.
 - iii. Preview the data and click Load.

3. Clean and Transform Data

- a. Select Home → Transform Data to open Power Query Editor.
- b. Select the Sales table.
- c. Ensure that the Date column has the Date data type.
- d. Create a calculated column named TotalSales using:
- e. $\text{Quantity} \times \text{UnitPrice}$
- f. Ensure that Quantity, UnitPrice, and TotalSales have appropriate numeric data types.
- g. Select the Targets table and ensure that the Month column has the appropriate Date data type.
- h. Select Close & Apply to load the transformed data.

4. DAX Measures

- a. The following measures were created:
 - i. Total Sales
 - ii. `Total Sales = SUM(Sales[TotalSales])`

- iii. Total Quantity
- iv. $\text{Total Quantity} = \text{SUM}(\text{Sales}[\text{Quantity}])$
- v. Average Order Value
- vi. $\text{Avg Order Value} = \text{DIVIDE}([\text{Total Sales}], \text{COUNTROWS}(\text{Sales}))$
- vii. Target Sales
- viii. $\text{Target Sales} = \text{SUM}(\text{Targets}[\text{TargetSales}])$

5. **Create Basic Visualizations**

- a. Bar Chart – Sales by Category
 - i. Insert a bar chart.
 - ii. Add Category to the Y-axis.
 - iii. Add Total Sales to the X-axis.
 - iv. Enable data labels and sort the chart in descending order.
- b. Line Chart – Monthly Sales Trend
 - i. Insert a line chart.
 - ii. Add YearMonth to the X-axis.
 - iii. Add Total Sales to the Y-axis.
 - iv. Set the title as Monthly Sales Trend.
- c. Donut Chart – Sales by Region
 - i. Insert a donut chart.
 - ii. Add Region to the Legend.
 - iii. Add Total Sales to the Values.
 - iv. Set the title as Sales by Region.

6. **Create KPI Cards**

- a. Three card visuals were created for:
 - i. Total Sales
 - ii. Total Quantity
 - iii. Average Order Value

7. **Create Top Products Table**

- a. Insert a table visual
- b. Add Product, Total Sales, and Total Quantity.
- c. Apply a Top N filter with 10 as the value.
- d. Use Total Sales as the value for ranking.

8. **Add an Interactive Region Slicer**

- a. Insert a slicer visual.
- b. Add Region to the slicer.
- c. Use the slicer to filter the dashboard by East, North, South, or West.

9. **Create Sales vs Target Gauge**

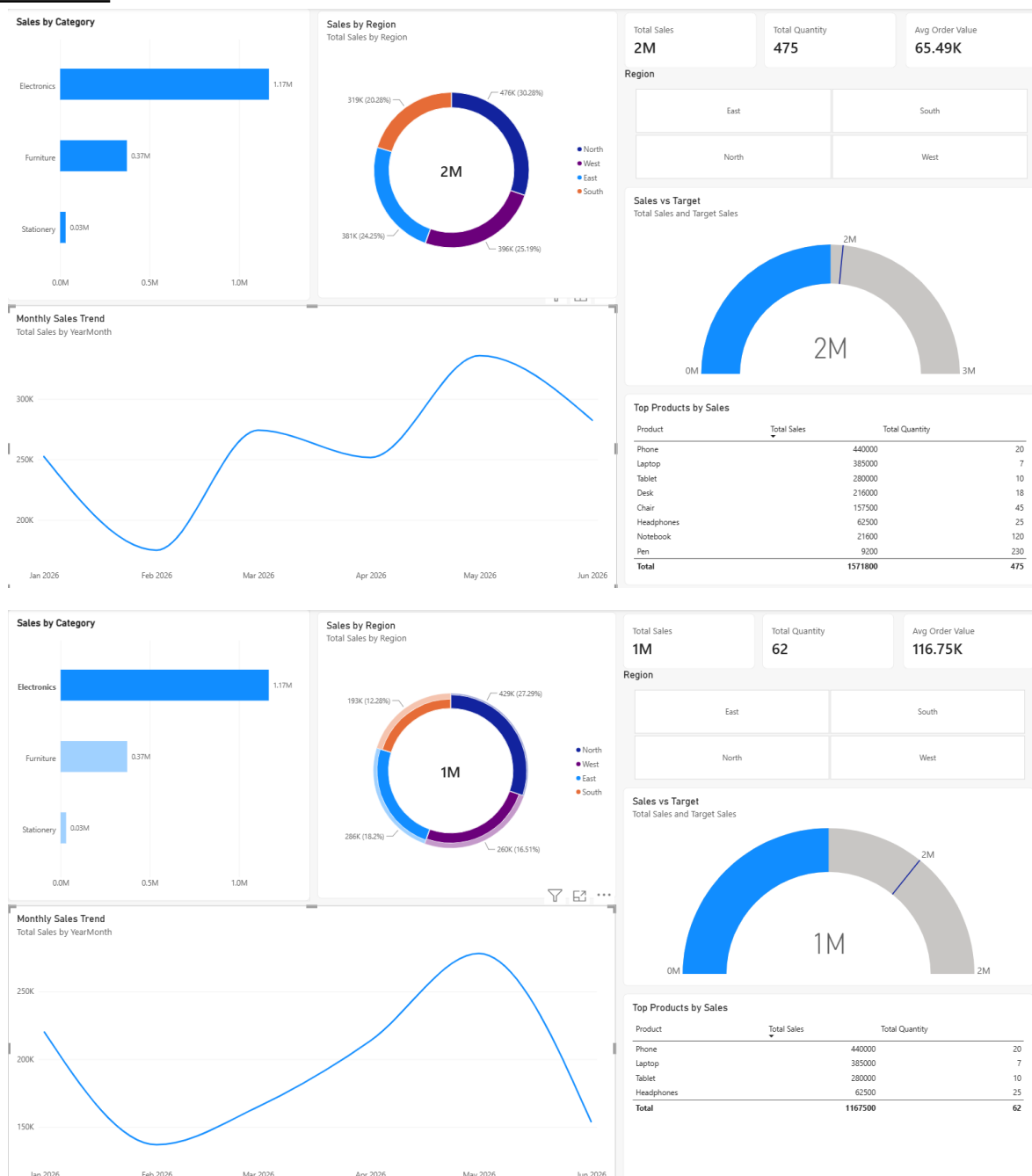
- a. Insert a gauge visual.

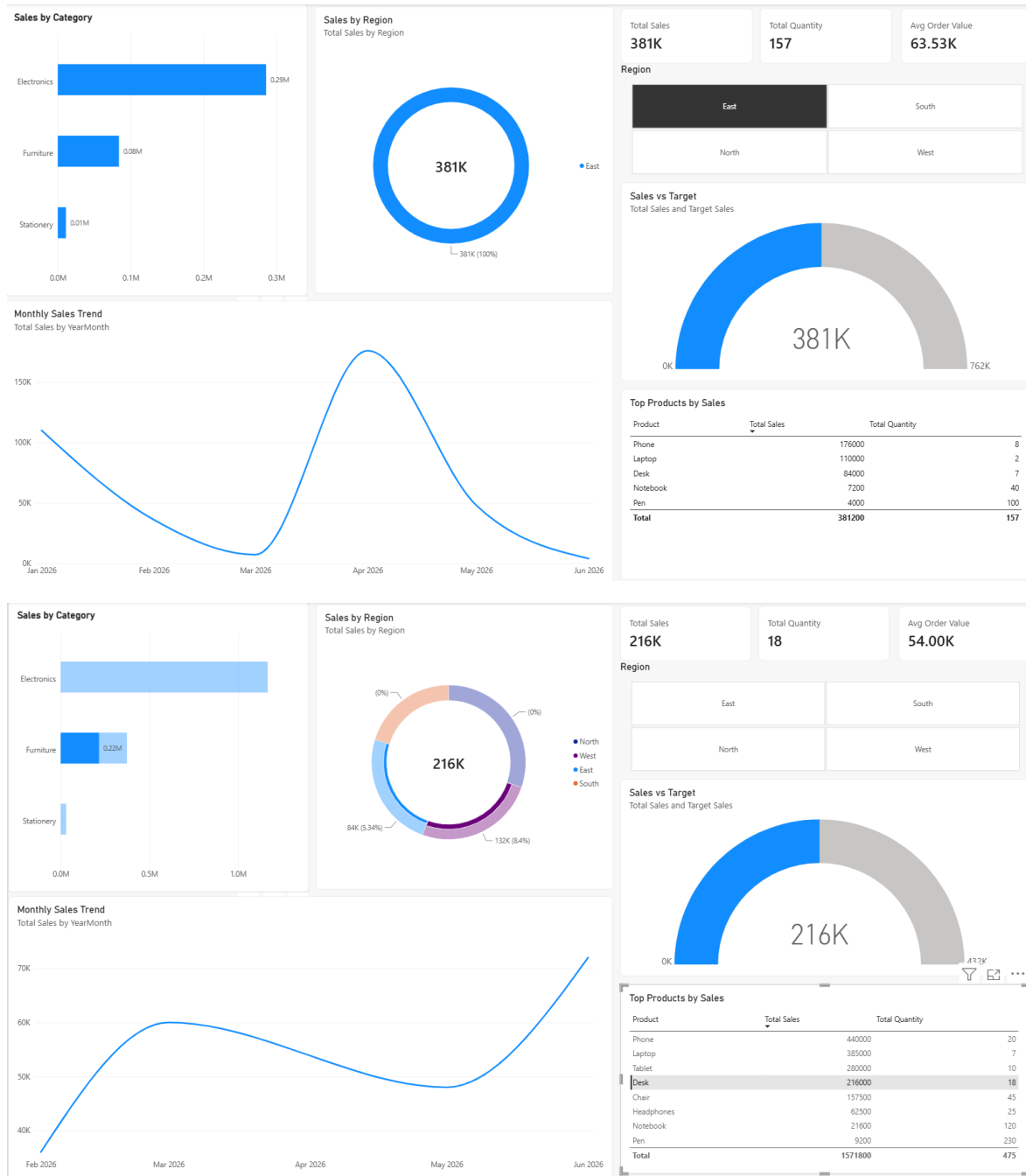
- b. Add Total Sales as the Value.
- c. Add Target Sales as the Target value.
- d. Set the title as Sales vs Target.

10. Assemble the Dashboard

- a. Arrange the KPI cards at the top.
- b. Place the category bar chart and region donut chart below the KPI cards.
- c. Place the sales-versus-target gauge beside the region slicer.
- d. Place the monthly sales trend and top-products table at the bottom.
- e. Apply suitable titles and formatting.
- f. Save the Power BI report as a .pbix file.

OUTPUT:





RESULT:

Thus, the given experiment was successfully performed using Power BI Desktop. The sales data was imported, transformed, analyzed using DAX, and presented through interactive visualizations and a dashboard.